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Project Title:

Multilingual Symptom Reasoning with Structured Risk Prediction for Female Reproductive Health

Project Area / Domain: Machine Learning and Artificial Intelligence

Project Aims & Objectives:

1. Develop a symptom-grounded multilingual dataset (FemSympQA) mapping patient-described symptoms to conditions, risk levels, and recommended actions using verified clinical sources.
2. Address the limitations of existing datasets lacking patient-style vocabulary and multilingual variation by generating diverse symptom expressions across languages.
3. Build a RAG-based system (HerHealthGPT) that retrieves evidence-backed medical knowledge for early-stage symptom interpretation.
4. Implement a risk prediction module and evaluate using measurable metrics such as classification accuracy and consistency across language variations.

Scope of Work:

In Scope:

- Construct dataset by extracting structured information from specific clinical web resources (listed below)
- Generate query variations using paraphrasing and multilingual translation (English, French, Arabic)
- Evaluate system against direct LLM baseline and non-multilingual baseline

Out of Scope:

- Real patient data collection
- Clinical trials
- Training large-scale medical foundation models

Expected Deliverables:

1. A curated dataset (FemSympQA) with multilingual symptom variations
2. A working HerHealthGPT prototype integrating RAG and agent-based reasoning

Prerequisite Knowledge / Skills for Student:

- Python programming
- Machine learning fundamentals
- Natural Language Processing and embeddings
- Familiarity with RAG and vector search
- Basic understanding of healthcare AI

Data / Resources Available:

Existing Datasets

- [A Women's Health Benchmark for Large Language Models](#)
- [Women Health Dataset Source](#)
- [Menstrual Health Awareness Dataset](#)

Primary Clinical Sources (Symptom → Risk → Action Extraction)

- Polycystic Ovary Syndrome (PCOS) – Symptoms and Causes
<https://www.nhs.uk/conditions/polycystic-ovary-syndrome-pcos/symptoms/>
- Heavy Periods (Menorrhagia)
<https://www.nhs.uk/conditions/heavy-periods/>
- Infertility – Symptoms and Causes (Female)
<https://www.nhs.uk/conditions/infertility/>
- PCOS – CDC Overview
<https://www.cdc.gov/diabetes/basics/pcos.html>
- Endometriosis – NIH Resource
<https://www.nichd.nih.gov/health/topics/endometri/conditioninfo>

Dataset Construction Strategy (Explicit and Reproducible)

The dataset will be constructed by:

1. Extracting symptoms, conditions, and recommended actions from NHS and CDC pages
2. Mapping them into structured format
3. Expanding each symptom into:
 - a. paraphrased queries
 - b. multilingual variants
4. Validating entries against multiple sources (NHS + CDC + NIH)

Tools and Models:

- Data Collection: Python (BeautifulSoup, requests), Pandas
- Text Processing: spaCy, Regex
- Paraphrasing: T5 (t5-base), PEGASUS
- Multilingual Translation: NLLB-200 (facebook/nllb-200-distilled-600M), mT5
- Embeddings: Sentence-BERT (all-MiniLM-L6-v2)
- Vector Retrieval: FAISS (or ChromaDB)
- RAG Framework: LangChain or LlamaIndex
- LLM for Reasoning: Mistral 7B / LLaMA 3 / GPT-4 (API)
- Risk Prediction: Scikit-learn (Logistic Regression, XGBoost) or PyTorch
- Recommendation Module: Rule-based logic / lightweight decision model
- Evaluation (NLP): BERTScore, ROUGE
- Evaluation (Classification): Accuracy, F1-score
- Development Environment: Google Colab, Jupyter Notebook
- Version Control: GitHub